# Federal Survey Question Consolidation

**DRAFT — not validated — see disclaimer slide**

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## Disclaimer

> Warning
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> **DRAFT** — These materials are in development and have not been independently validated. The author makes no claims regarding validity until formal review is complete.

The views expressed are the author's own and do not necessarily represent the views of the U.S. Census Bureau or the U.S. Department of Commerce.

This is exploratory research assessing feasibility of AI-assisted survey analysis methods.

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## The Problem

- Federal surveys ask overlapping questions
- Respondent burden + data silos
- **This report:** Capstone analysis — which questions can actually consolidate?

Brief — already covered in Reports 1-2

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## Why It Matters

**For Respondents:** - Declining survey response rates + increasing costs - Overlapping questions create unnecessary burden - Consolidation reduces redundant questions

**For Federal Agencies:** - **Expert time is the scarce resource**, not expert capability - Traditional approach: weeks/months of manual comparison - **Our goal:** Multiply expert effectiveness → faster burden reduction

Dual value proposition: (1) Burden reduction for respondents (2) Expert efficiency to achieve it faster

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## The Challenge

- Determining consolidation **requires expert judgment** (this doesn't change)
- Traditional timeline: **weeks to months** for manual review of 1,598 pairs
- Our approach: **AI handles pre-work, experts focus on highest-value decisions**
- Result: Structured expert review table ready for validation

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## What Experts Get

| What AI Provides | What Experts Do |
|---|---|
| Pre-screened pairs with best matches | Validate match quality |
| Initial F1/F2/F3 classification | Confirm or override |
| Documented reasoning | Understand and critique |
| Confidence scores | Prioritize review effort |
| Barrier codes | Verify diagnosis |

**76%** classified with high confidence → spot-check **24%** flagged for expert review → where judgment adds most value

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### Starting Point

- **7,400+** survey questions across **49** federal surveys
- Mapped to Census Bureau standard taxonomy
- [Report 1 established concept classification]

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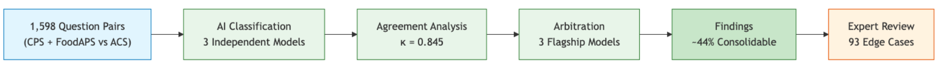
### Concept Classification (Report 1)

- LLM ensemble with arbitrator pattern
- Each question → best concept match
- Taxonomy: Demographics, Economics, Education, Health, etc.

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### Key Insight

> **Important**
>
> Same concept   swappable data

- "Income" questions can measure completely different constructs
- Demographics (age, sex, race) = expected overlap (minority of questions)
- **Specialized content** = where real analysis matters

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### Harmonization Framework

Adopted standard survey harmonization classification:

| Level | Definition |
|---|---|
| **F1** | Directly consolidable — no transformation needed |
| **F2** | Consolidable with transformation — needs recoding |
| **F3** | Not consolidable — fundamental barriers |

| Level | Definition |
|---|---|

Barrier codes: CC (construct/concept), TC (temporal), RS (response scale)

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## Pairwise Comparison

- Within concept categories, compare all question pairs
- **1,598 pairs** analyzed (CPS-ACS, FoodAPS-ACS)
- Naive but exhaustive — expect many F3, looking for matches

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## Multi-Model Ensemble

Three frontier models to reduce bias:

| Model | Behavior |
|---|---|
| OpenAI | Moderate synthesis, slight pro-self bias |
| Anthropic | High synthesis, neutral |
| Google | Deferential, conservative |

Different profiles = documented finding, not noise

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## Agreement & Arbitration

- **= 0.845** — high inter-rater agreement
- Voting, entropy, Borda scoring
- Arbitration resolves disagreements → final verdict

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## Evaluating AI for Survey Harmonization

**Construct Validity:** - Three different architectures (OpenAI, Anthropic, Google) with different training - **High convergence ( = 0.845)** = task is well-defined, not model artifact

**Documented Behavioral Differences:** - Google: Deferential, conservative (low synthesis) - OpenAI: Moderate synthesis, slight self-bias - Anthropic: High synthesis, neutral

**Cost/Quality Design:** Fast models rate → flagship models arbitrate

**Result:** Reusable methodology for future survey harmonization work

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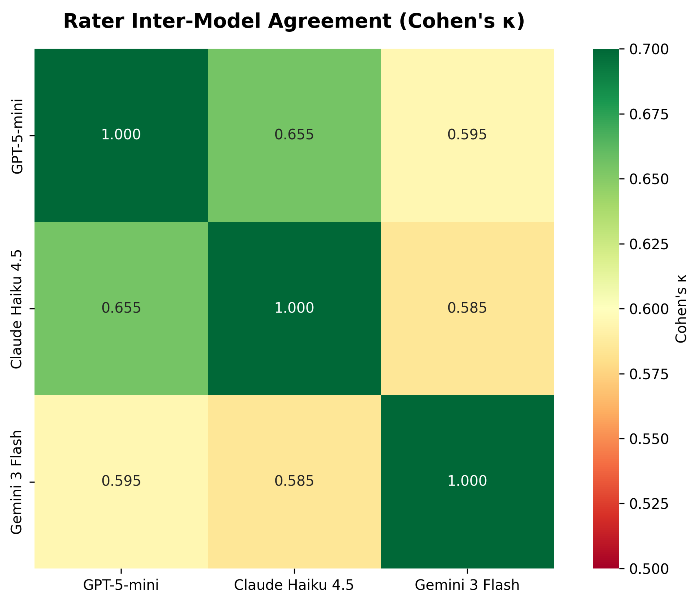
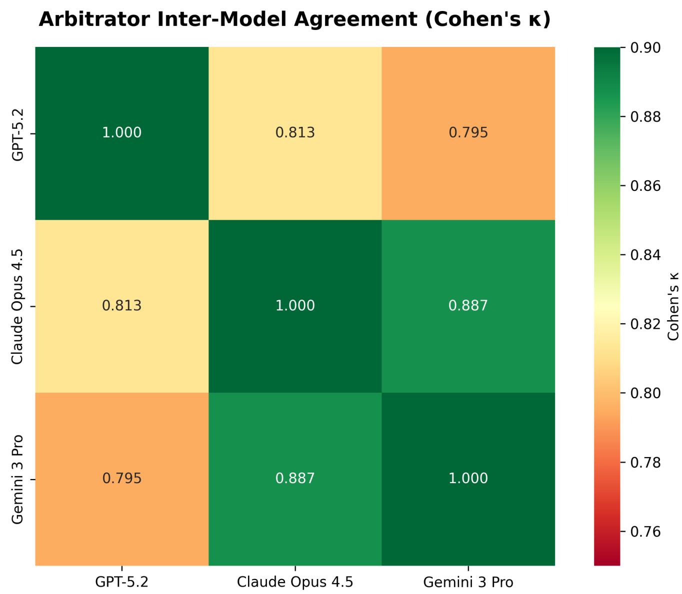
## Construct Validity: Why Trust These Results?

Three independent architectures → same judgments = well-defined task




### Rater Agreement

Fleiss = 0.611 (substantial)




## Arbitrator Agreement

Fleiss $= 0.843$ (almost perfect)

Convergence INCREASES with reasoning depth ($0.611 \rightarrow 0.843$)

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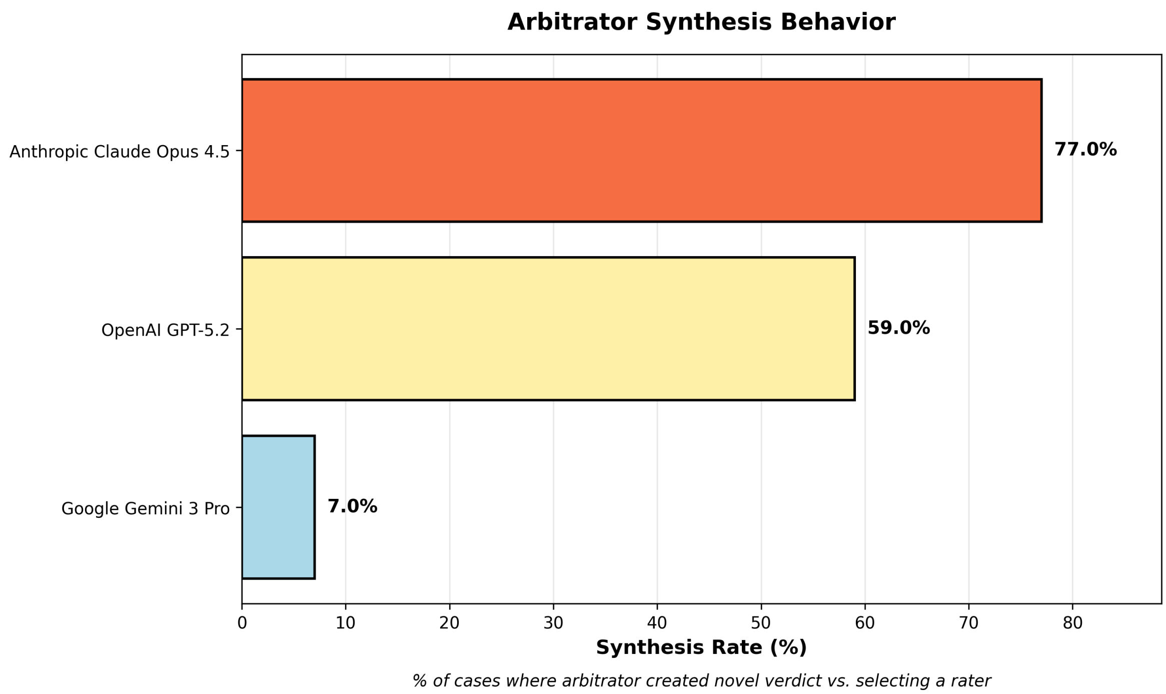
## Arbitrator Behavioral Differences

Not a bug — a feature. Diversity captures multiple valid interpretations.




| Arbitrator | Synthesis Rate | Style |
|---|---|---|
| Google (Gemini 3 Pro) | 7% | Deferential — trusts raters |
| OpenAI (GPT-5.2) | 59% | Balanced |
| Anthropic (Claude Opus 4.5) | 77% | Active synthesis |

No single model dominates. Ensemble balances conservative and synthetic approaches.

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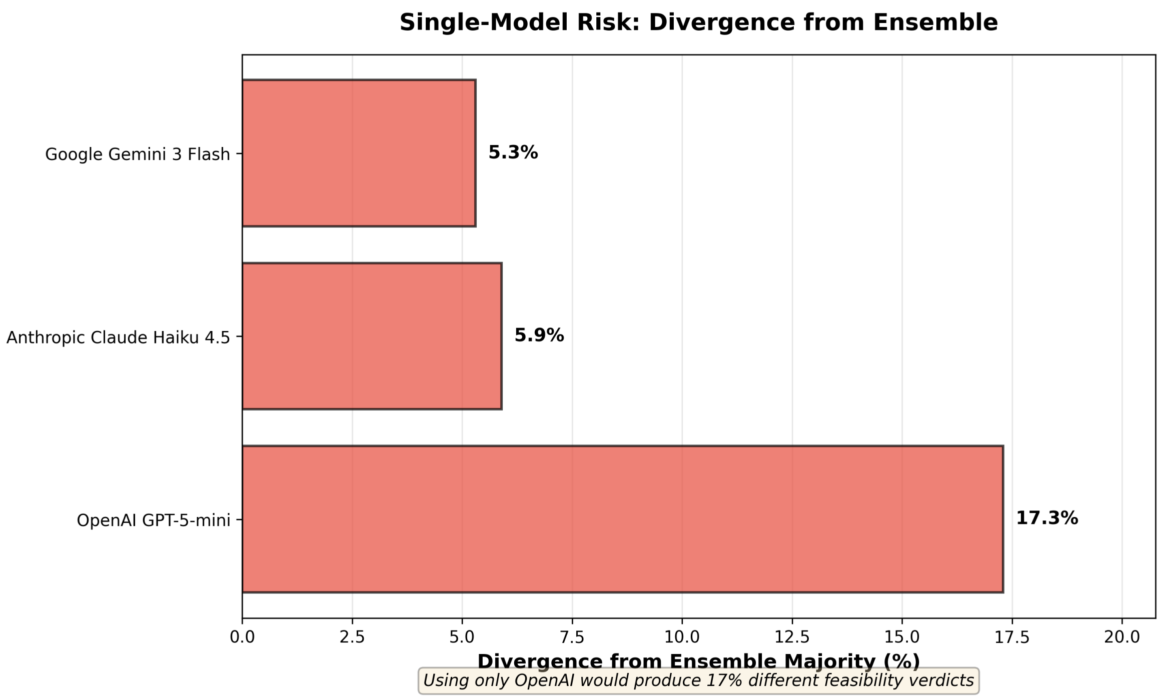
## Why Not Just Use One Model?

Single-model divergence from ensemble majority:




- **OpenAI alone:** 17.3% different feasibility verdicts (~277 pairs)
- **Anthropic alone:** 5.9% (~94 pairs)
- **Google alone:** 5.3% (~85 pairs)

The 3rd model catches what any 2 would miss.

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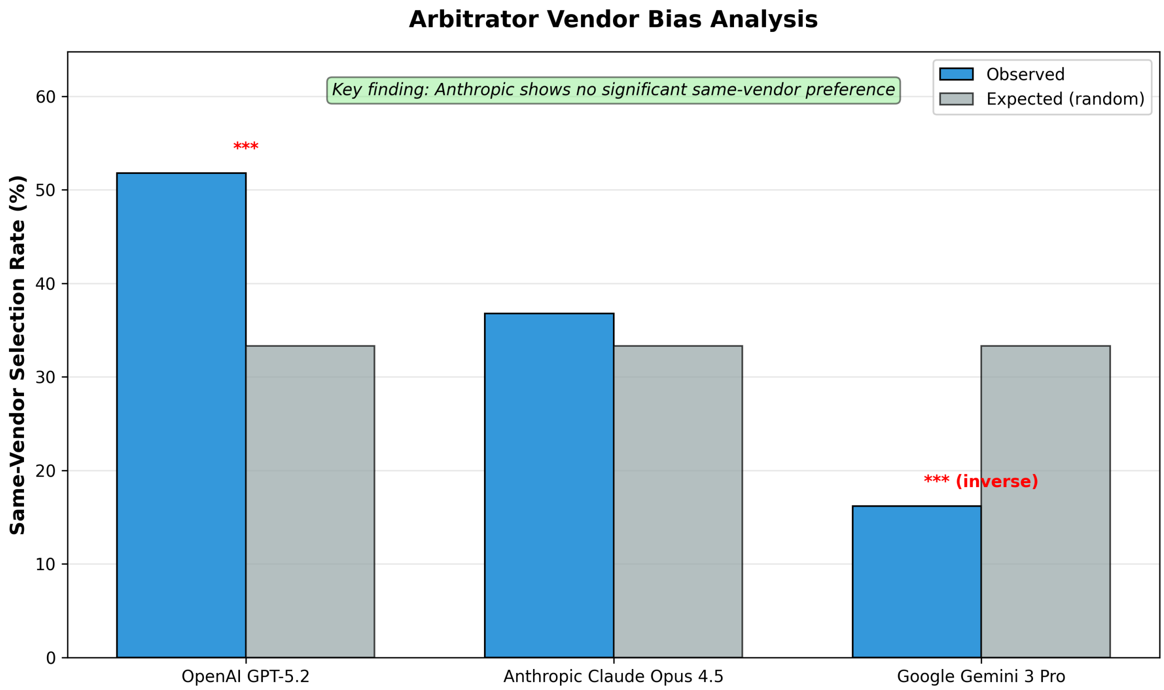
## Do Arbitrators Favor Their Own Vendor's Rater?




| Arbitrator | Same-Vendor Selection | Expected | Significant? |
|---|---|---|---|
| OpenAI | 51.8% | 33.3% | Yes ($p<0.001$) |
| Anthropic | 36.8% | 33.3% | **No** ($p=0.159$) |
| Google | 16.2% | 33.3% | Yes — AGAINST own |

**Finding:** Anthropic shows no same-vendor preference. Google actively avoids own vendor.

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## Methodology Validation Summary

| Validation Check | Result | Interpretation |
|---|---|---|
| Rater convergence | = 0.611 | Substantial — task well-defined |
| Arbitrator convergence | = 0.843 | Almost perfect — disagreements resolvable |
| Single-model risk | 5-17% | Multi-model approach justified |

| Validation Check | Result | Interpretation |
|---|---|---|
| Vendor bias | Mixed | Anthropic unbiased; OpenAI self-favoring |

**Conclusion:** Multi-model ensemble provides robust classifications suitable for survey harmonization analysis.

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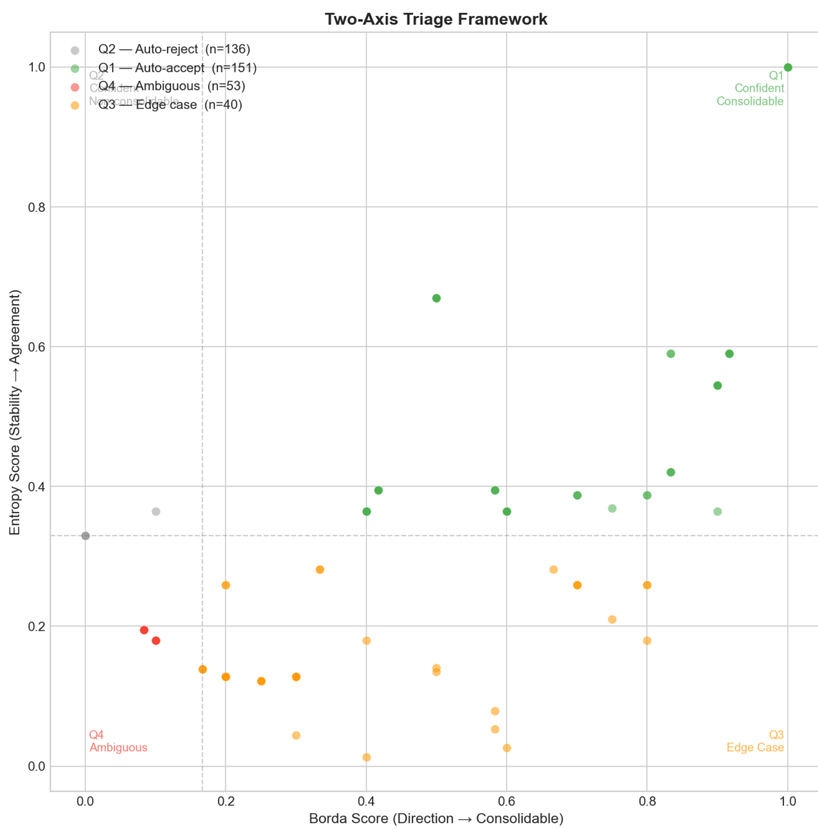
## Question-Level Rollup

- Pair analysis → **380 unique source questions**
- Each with best ACS match
- Two-axis triage: Direction (Borda) × Stability (Entropy)

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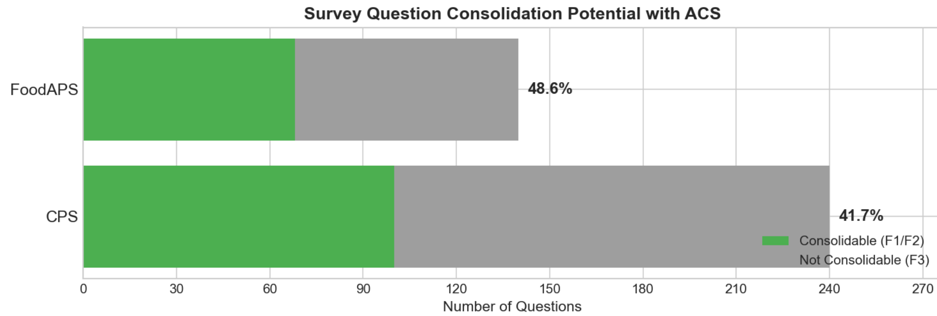
## Headline Results




- **CPS:** 41.7% consolidable (100 of 240 questions)
- **FoodAPS:** 48.6% consolidable (68 of 140 questions)

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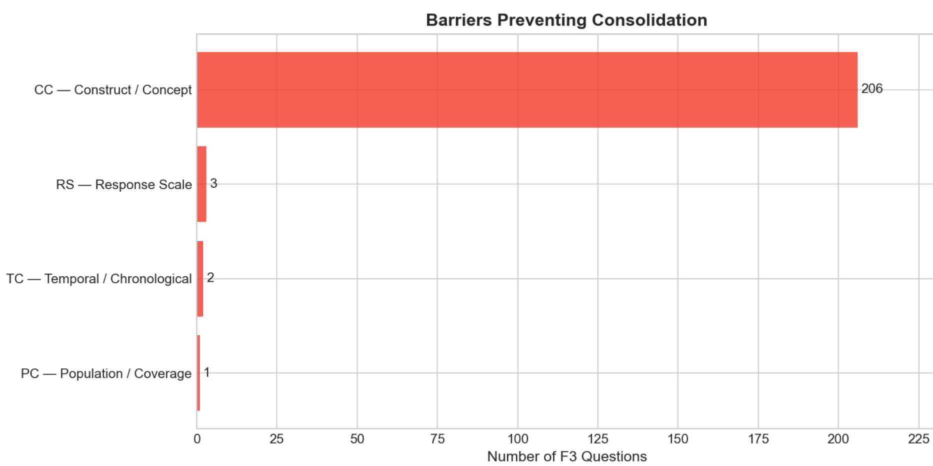
## Why Questions Can't Consolidate




**97% of F3 = Construct/Concept differences (CC)**

Not fixable barriers — fundamentally different measurements

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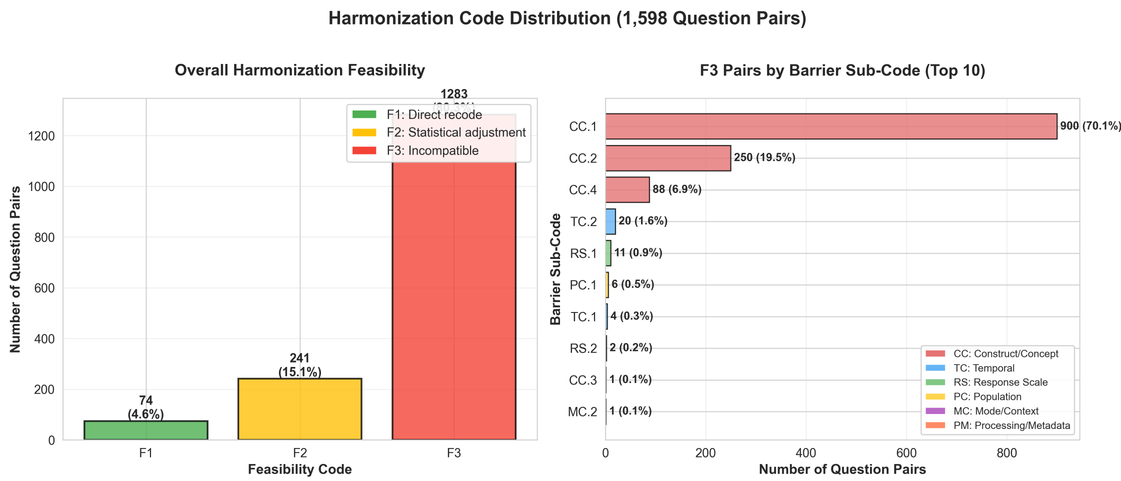
## Harmonization Code Distribution




Left: Overall F1/F2/F3 distribution across 1,598 pairs Right: Top 10 barrier sub-codes for F3 pairs - CC.1 (concept definition) dominates at 70%

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## Example: High Consolidability (F1)

**CPS:** "How many hours per week do you USUALLY work?"

**ACS:** "What is your best estimate of hours per week you usually work?"

- Verdict: **F1** (directly consolidable)
- Confidence: Borda=1.0, Entropy=1.0
- Same construct, same framing, same reference

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## Example: Medium Consolidability (F2)

**CPS:** "EXCLUDING overtime, tips, commissions — what is hourly rate?"

**ACS:** "What is your hourly rate of pay?"

- Verdict: **F2** (consolidable with transformation)
- Transformation needed: scope alignment (ACS includes all components)

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### Example: Not Consolidable (F3)

**CPS:** "Do you WANT to work full time (35+ hours)?"

**ACS:** "Did this person WORK for pay last week?"

- Verdict: **F3** — Barrier: **CC.1**
- Same domain (employment), entirely different constructs
- Preference vs. behavior — not reconcilable

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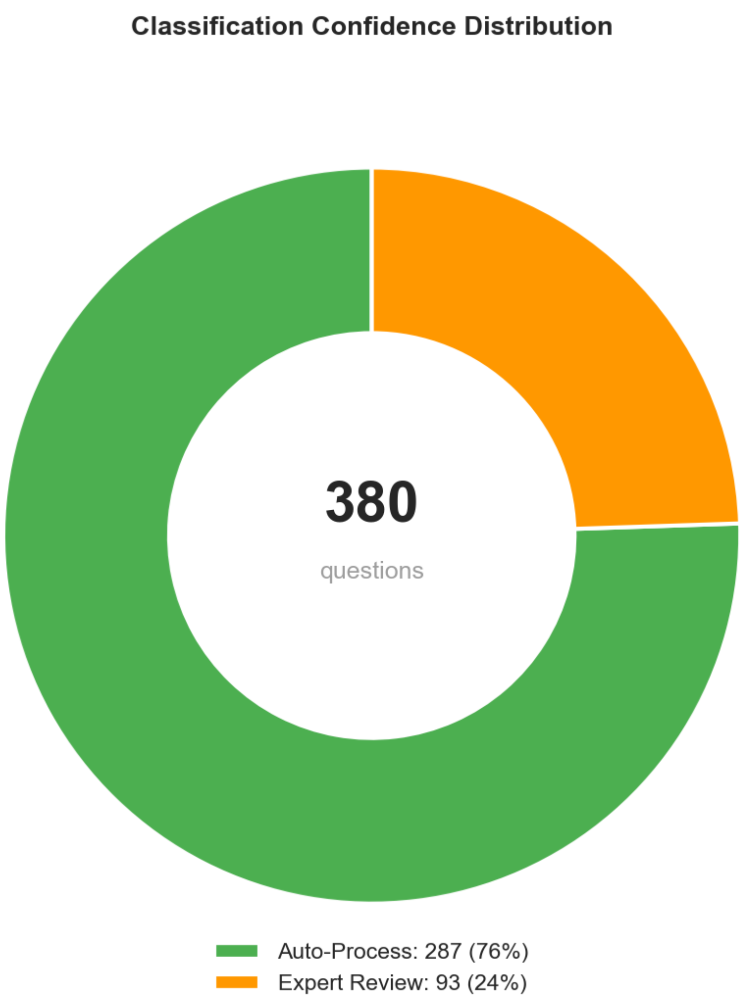
### Expert Review Load



- **76%** (287 questions) classified with high confidence → spot-check recommended
- **24%** (93 questions) flagged for expert review → **where human judgment adds most value**
- **This IS the efficiency gain:** Experts review 93 questions, not 1,598 pairs

Traditional: Expert reviews all 1,598 pairs = weeks/months AI-assisted: Expert reviews 93 flagged questions = focused effort on highest-ROI decisions

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## Deliverables

**Primary:** Expert review table (`expert_review_combined.csv`) - 380 questions with best ACS matches - F1/F2/F3 classifications with confidence scores - Barrier codes and reasoning for each decision - Triage routing (Q1-Q4) for prioritized review

**Breakdown:** - 168 consolidable (F1+F2) — ready for validation - 212 non-consolidable (F3) — with documented barriers - 93 flagged for expert review — highest-value decisions - Full methodology — reproducible for additional surveys

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## What This Proves

- **< 1 day processing time** for work that traditionally takes weeks/months
- **Expert efficiency → faster burden reduction** — identify consolidation opportunities at scale
- **Human oversight preserved throughout** — AI provides initial judgment, experts validate
- **The table is the deliverable** — 380 questions, pre-analyzed, ready for expert review

AI doesn't replace burden reduction goals — it accelerates the path to achieving them through expert efficiency

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## What's Next

1. Expert validation of recommendations
2. Expand to additional federal surveys
3. Application to survey editing/imputation

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## Summary

**Findings:** - ~**44% consolidation potential** — 168 questions could reduce respondent burden - **97% of failures** = construct differences (not fixable)

**Method: - < 1 day processing** for work that traditionally takes weeks/months - **AI provides initial judgment, experts provide final validation - The real product:** 380 questions, pre-analyzed, ready for expert review

**Impact:** Accelerates path to burden reduction through expert efficiency

Expert review table: output/analysis/expert_review_combined.csv Dual value: (1) Identifies burden reduction opportunities (2) Efficiently, preserving expert judgment

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## Questions?

Contact: [your info]

Repository: [link]

Expert review tables: [link to output]

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# Appendix

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### Cost/Quality: Is the 3rd Model Worth It?

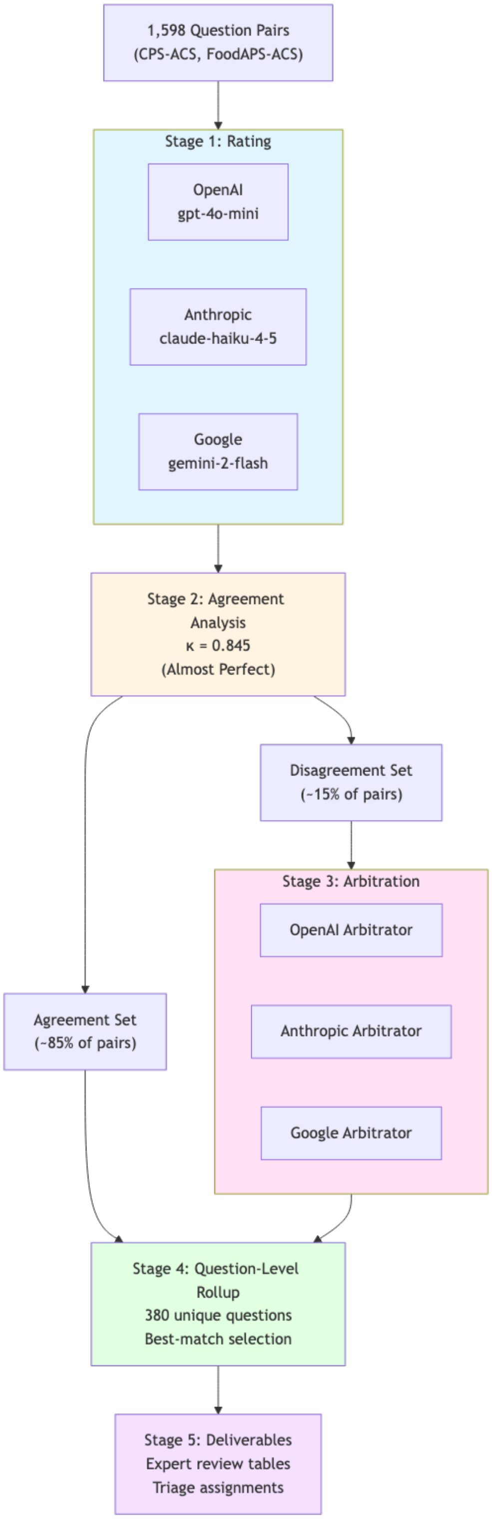
**Two-model scenarios — tie frequency:**

| Pair | Feasibility Ties |
|---|---|
| OpenAI + Anthropic | 23.2% |
| OpenAI + Google | 22.1% |
| Anthropic + Google | 11.0% ← lowest |

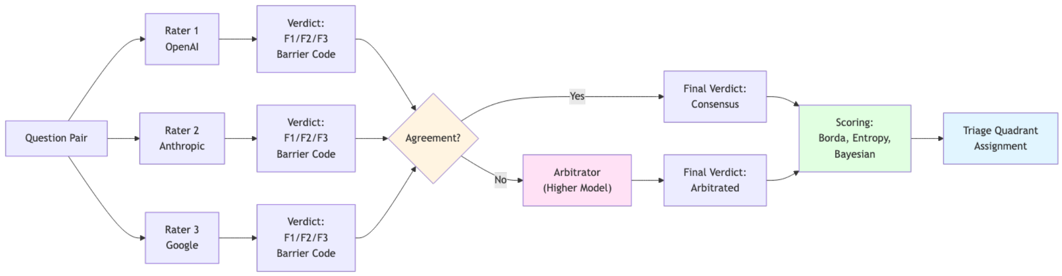
**Recommendation:** - <1,000 pairs: 2 models + human tiebreaking - >1,000 pairs: 3 models for automated triage

The 3rd model's value is highest when first two disagree.

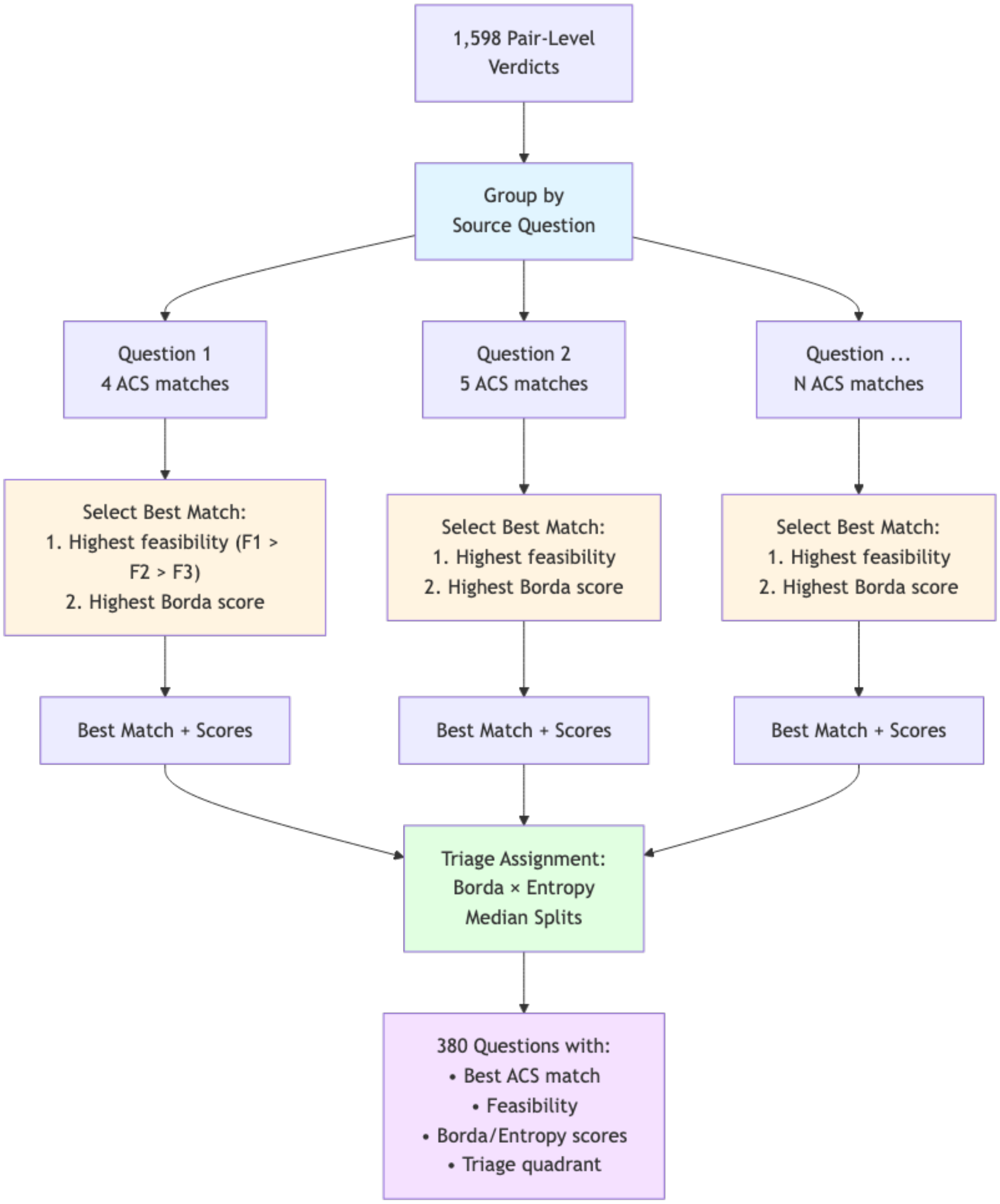
## Pipeline Architecture



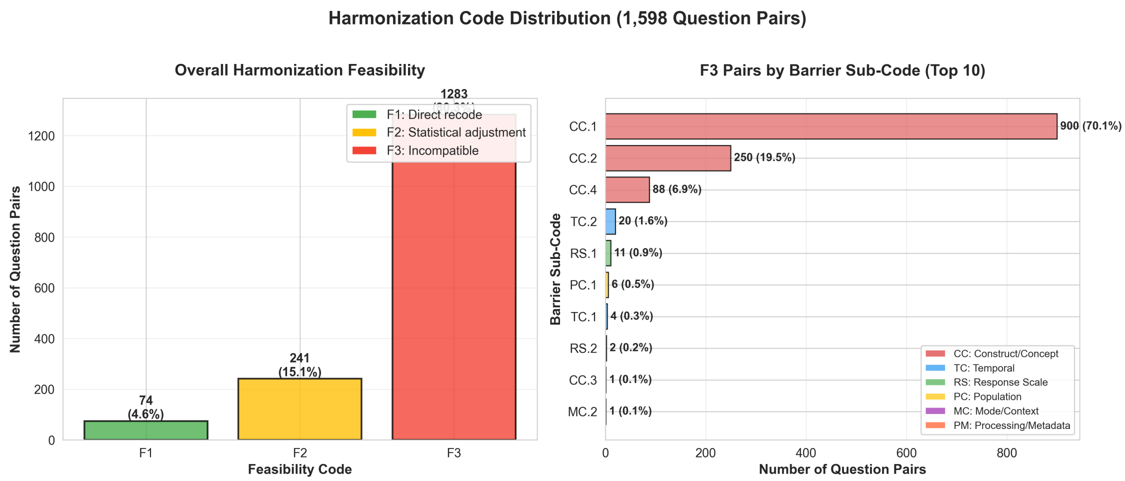
## LLM Ensemble Pattern



### Question-Level Rollup



## Harmonization Code Distribution




Left: Overall F1/F2/F3 distribution across 1,598 pairs Right: Top 10 barrier sub-codes for F3 pairs - CC.1 (concept definition) dominates at 70%

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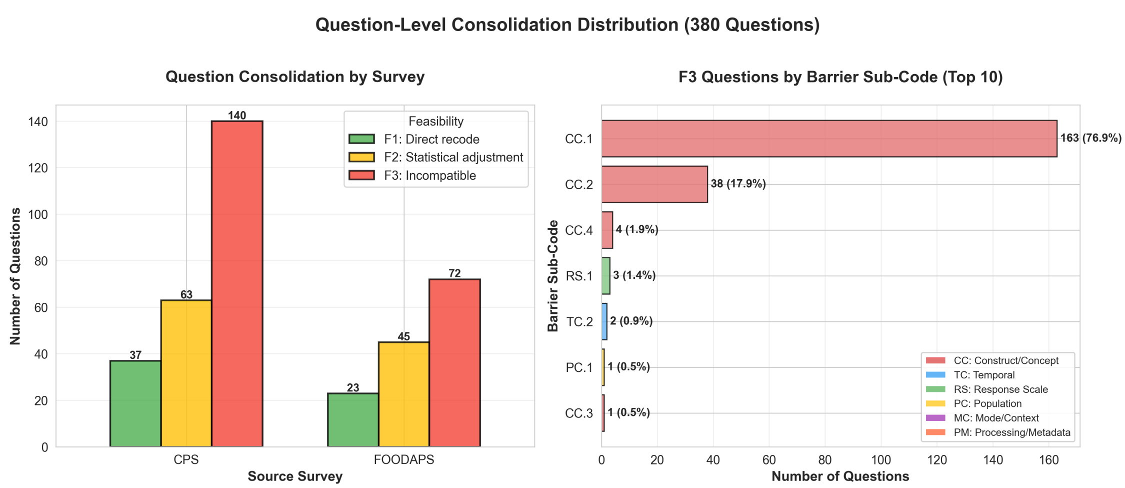
## Question-Level Consolidation Distribution




Left: CPS (41.7% consolidable) vs FOODAPS (48.6% consolidable) breakdown Right: F3 questions by barrier code - CC.1 dominates at 76.9% of non-consolidable questions

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## Harmonization Taxonomy

**Primary Source:** Fortier et al. (2011, 2017) DataSHaPER/Maelstrom framework

**Supporting:** Wolf et al. (2016), Saris & Gallhofer (2014), Slomczynski & Tomescu-Dubrow (2018)

| Code | Feasibility | Definition | Action Required |
|---|---|---|---|
| **F1** | **Direct recode** | Mechanically transformable with simple recoding or collapsing | Simple data transformation |
| **F2** | **Statistical adjustment** | Requires modeling, imputation, or bridging studies | Statistical harmonization |
| **F3** | **Incompatible** | Fundamentally different constructs; not harmonizable without re-fielding | No harmonization possible |

Fortier et al. (2011) found 38% of variables could be harmonized across 53 studies

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## Barrier Code Taxonomy (1/3)

### Temporal Constraints (TC)

| Code | Subtype | Definition | Example |
|---|---|---|---|
| **TC** | Temporal/Chronological | Reference period or timing differences | — |
| TC.1 | Reference period length | Different recall windows | 7-day vs 12-month |
| TC.2 | Temporal framing | Point-in-time vs habitual vs retrospective | “last week” vs “usually” |
| TC.3 | Calendar alignment | Fixed vs rolling reference periods | Calendar year vs “past 12 months” |

### Construct Constraints (CC)

| Code | Subtype | Definition | Example |
|---|---|---|---|
| **CC** | Construct/Concept | Concept definition or operationalization differences | — |
| CC.1 | Concept definition | Different meaning of core term | “Employment” including/excluding unpaid work |
| CC.2 | Operationalization | Different behavioral indicators | “Food insecurity” via different batteries |
| CC.3 | Boundary conditions | Different thresholds or cutoffs | Income brackets with different boundaries |
| CC.4 | Scope inclusions | Different components counted | “Income” including/excluding specific sources |

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## Barrier Code Taxonomy (2/3)

### Population/Coverage Constraints (PC)

| Code | Subtype | Definition | Example |
|---|---|---|---|
| **PC** | Population/Coverage | Universe, frame, or sample design differences | — |
| PC.1 | Universe definition | Target population differs | Civilian vs all residents |
| PC.2 | Frame exclusions | Different sampling exclusions | Group quarters, territories |
| PC.3 | Age bounds | Different age eligibility | 15+ vs 16+ vs 18+ |
| PC.4 | Geographic scope | Different geographic coverage | 50 states vs states + territories |

### Response Scale Constraints (RS)

| Code | Subtype | Definition | Example |
| --- | --- | --- | --- |
| **RS** | Response Scale | Scale type, categories, or format differences | — |
| RS.1 | Scale type | Fundamentally different formats | Binary vs 5-point vs continuous |
| RS.2 | Category structure | Different number/boundaries | 5-point vs 7-point Likert |
| RS.3 | Anchoring/labels | Different verbal anchors | "Strongly agree" differences |
| RS.4 | Numeric vs verbal | Number scale vs labeled categories | 0-10 slider vs frequency labels |

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## Barrier Code Taxonomy (3/3)

### Mode/Context Constraints (MC)

| Code | Subtype | Definition | Example |
| --- | --- | --- | --- |
| **MC** | Mode/Context | Interview mode or questionnaire context differences | — |
| MC.1 | Interview mode | Different data collection modes | CATI vs web self-administered |
| MC.2 | Question routing | Different skip patterns | Subset vs all respondents |
| MC.3 | Contextual priming | Preceding questions affect interpretation | Question order effects |
| MC.4 | Proxy response | Proxy vs self-report rules | Household head vs individual |

### Processing/Metadata Constraints (PM)

| Code | Subtype | Definition | Example |
| --- | --- | --- | --- |
| **PM** | Processing/Metadata | Coding, weighting, or documentation differences | — |

| Code | Subtype | Definition | Example |
|---|---|---|---|
| PM.1 | Coding schemes | Different classification systems | Occupation coding differences |
| PM.2 | Derived variables | Different construction algorithms | Poverty calculation methods |
| PM.3 | Documentation gaps | Insufficient metadata | Missing questionnaire context |

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## Triage Quadrants: Operational Framework

Two-axis classification using Borda (direction) and Entropy (stability):

| Quadrant | Borda | Entropy | Count | Description | Action |
|---|---|---|---|---|---|
| **Q1** | High | High | 151 | Confident consolidable — classifiers agreed, leaning accept | Auto-process, spot-check |
| **Q2** | Low | High | 136 | Confident non-consolidable — classifiers agreed, leaning reject | Auto-process, low priority |
| **Q3** | High | Low | 40 | Uncertain accept — leaning yes but contested | **Expert review priority** |
| **Q4** | Low | Low | 53 | Uncertain reject — genuinely ambiguous | Expert review secondary |

**Thresholds:** Median split on question-level best-match scores

- Borda median: 0.167
- Entropy median: 0.330

**Rationale:** Separates "what's the answer?" (Borda) from "how much did they argue?" (Entropy)

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## Key Citations


**Harmonization Framework:**

- Fortier, I., et al. (2011). Is rigorous retrospective harmonization possible? Application of the DataSHaPER approach across 53 large studies. *International Journal of Epidemiology*, 40(5), 1314-1328.
- Fortier, I., et al. (2017). Maelstrom Research guidelines for rigorous retrospective data harmonization. *International Journal of Epidemiology*, 46(1), 103-115.

**Survey Methodology:**

- Wolf, C., et al. (2016). Harmonizing survey questions between cultures and over time. In *The SAGE Handbook of Survey Methodology* (pp. 502-524). SAGE.
- Saris, W. E., & Gallhofer, I. N. (2014). *Design, Evaluation, and Analysis of Questionnaires for Survey Research* (2nd ed.). Wiley.

**Data Harmonization:**

- Slomczynski, K. M., & Tomescu-Dubrow, I. (2018). Basic principles of survey data recycling. In *Advances in Comparative Survey Methods* (pp. 937-962). Wiley.
- Tomescu-Dubrow, I., et al. (Eds.). (2024). *Survey Data Harmonization in the Social Sciences*. Wiley.